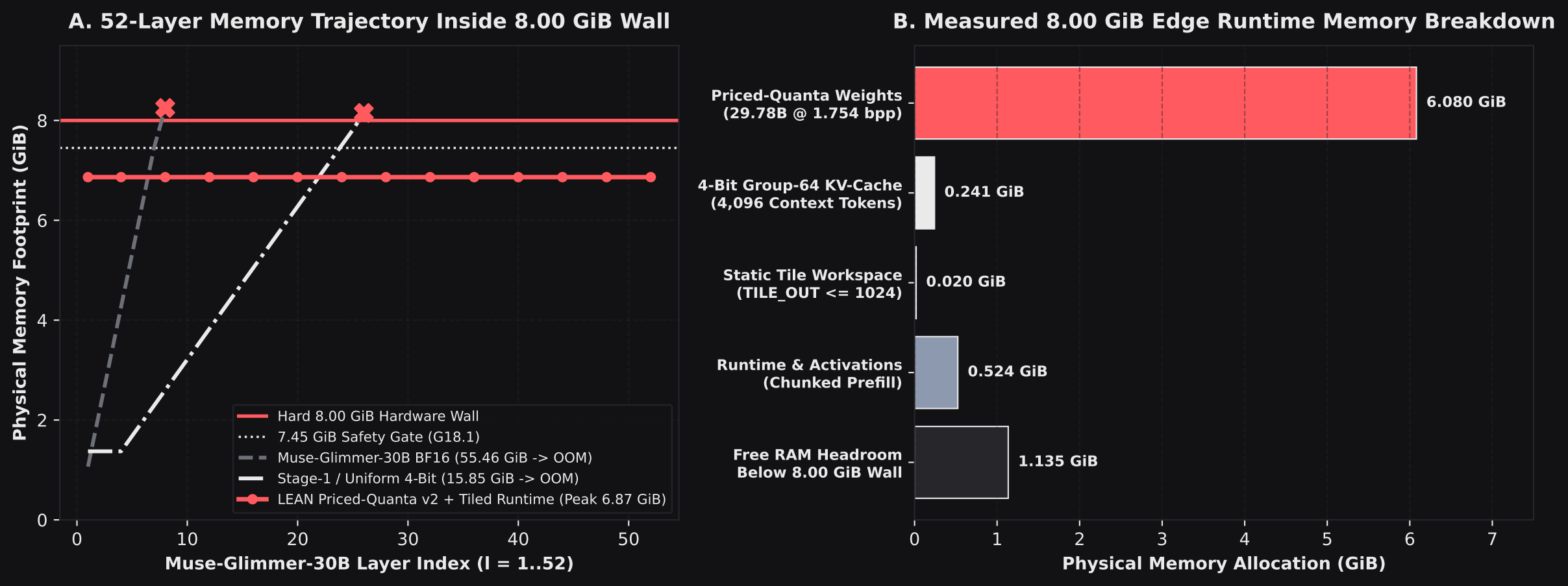
A. 52-Layer Memory Trajectory Inside 8.00 GiB Wall
Physical Memory Footprint (GiB)
0
2
4
6
8
0
10
20
30
40
50
Muse-Glimmer-30B Layer Index (l = 1..52)
Hard 8.00 GiB Hardware Wall
7.45 GiB Safety Gate (G18.1)
Muse-Glimmer-30B BF16 (55.46 GiB -> OOM)
Stage-1 / Uniform 4-Bit (15.85 GiB -> OOM)
LEAN Priced-Quanta v2 + Tiled Runtime (Peak 6.87 GiB)
B. Measured 8.00 GiB Edge Runtime Memory Breakdown
Priced-Quanta Weights
(29.78B @ 1.754 bpp)
6.080 GiB
4-Bit Group-64 KV-Cache
(4,096 Context Tokens)
0.241 GiB
Static Tile Workspace
(TILE_OUT <= 1024)
0.020 GiB
Runtime & Activations
(Chunked Prefill)
0.524 GiB
Free RAM Headroom
Below 8.00 GiB Wall
1.135 GiB
0
1
2
3
4
5
6
7
Physical Memory Allocation (GiB)